Set Up a Certificate Authority in RHEL5

Security certificates are widely used for authentication. This article explores how to set up a Certificate Authority in RHEL5.

Security certificates are basically used for authentication purposes and you must have encountered a number of websites that use them. These digital certificates are issued by a Certificate Authority. Such certificates contain the public key of the applicants and various other information regarding their identity. In this article, we'll discuss the setting up of a Certificate Authority in RHEL and certificate signing.

Before we start, just make sure you have *openssl* installed on your system and follow the steps listed below:

1. Open the */etc/pki/tls/openssl.cnf* file in a text editor and write down the following lines under the [CA_default] section:

```
dir=/etc/pki/myCA
certificate=$dir/my-ca.crt
crl=$dir/my-ca.crl
private_key=$dir/private/my-ca.key
```

You can find the purpose of each of these objects in the */etc/pki/tls/openssl.cnf* file.

2. Under the *[req_distinguished_name]* section, you can specify the default values for several fields:

```
countryName_default=IN
stateOrProvinceName_default=West Bengal
localityName_default=Kolkata
```

These fields are used during the time of certificate creation.

3. Now it is time to create your working directories using the following command:

```
mkdir -p /etc/pki/myCA/{certs,crl,newcerts,private}
```

4. Create a certificate index file using the command that follows:

```
touch /etc/pki/CA/index.txt
```

5. To create another file for the next certificate serial number to be issued, use the following command:

```
echo 01 > /etc/pki/myCA/serial
```

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